

Execution-order comparison

Pipeline: Normal LLM | 2 runs | 2 identical question texts | Reference: R1

Run	Folder / model
R1	exp1_llm_only_2026-10-09_19-37-53 llama3
R2	exp1_llm_only_2026-10-09_19-42-08 llama3

Main observations

These are observed differences, not a causal estimate of the effect of shuffling. Output length, background activity and measurement variation may contribute.

Recorded model names match; this does not establish identical model versions or generation settings.

Generation settings, token counts and RAG index identity are not consistently recorded by these reports; equality of those conditions cannot be verified.

Lowest observed emissions: R2, 1.1159 g CO₂e. Spread between highest and lowest: 0.12103 g (10.846% relative to the lowest).

R2 versus R1: identical execution sequence. 2 valid successful question pairs; 2 lower, 0 higher, 0 unchanged emissions.

Largest absolute change: Q1, -0.066311 g; execution position 1 to 1. Net change over these matched successful pairs: -0.12103 g.

R2: energy_wh change summed over 2 matched successful pairs: -0.16964.

R2: latency_s change summed over 2 matched successful pairs: -16.934.

Recorded metadata differences: Ordering, Seed, Runtime (s). Unknown or unrecorded values do not establish equivalence.

Historical zero defaults cannot be distinguished from measured zeros. Missing measurements are excluded and partial sums labelled. Percentage changes with a zero reference are N/A.

Run cost and coverage

Observed sums include failed requests. Coverage is valid readings / all questions. Partial sums cannot be treated as complete batch costs.

Run	Emissions (g)	Energy (Wh)	Request time (s)	Failures / unknown
R1	1.2369 [2/2]	1.7337 [2/2]	208.42 [2/2]	0
R2	1.1159 [2/2]	1.5641 [2/2]	191.49 [2/2]	0

Batch costs: R1-R2

Emissions (g CO2e)

Observed sums across all requests; partial coverage is labelled.

R1		1.237
	Valid 2/2	
R2		1.116
	Valid 2/2	

Energy (Wh)

Observed sums across all requests; partial coverage is labelled.

R1		1.734
	Valid 2/2	
R2		1.564
	Valid 2/2	

Metadata differences

Different recorded values are marked *. Not recorded means unavailable, not equal.

Setting	R1	R2
Folder *	exp1_llm_only_2026-10-09_19-37-53	exp1_llm_only_2026-10-09_19-42-08
Model(s)	llama3	llama3
Pipeline	Normal LLM	Normal LLM
Ordering *	Keep original order	Shuffle question types; keep questions within each question type in original order
Seed *	Not recorded	182766017
Questions	2	2
Runtime (s) *	208.71036314964294	191.53168296813965
Completion evidence	Summary present	Summary present
codecarbon_version	"3.3.0"	"3.3.0"
os	"Windows-11-10.0.26200-SP0"	"Windows-11-10.0.26200-SP0"
effective_scope	"machine"	"machine"
measure_power_secs	1	1
pue	1.0	1.0
country_iso_code	"IND"	"IND"
config_profile	"inference"	"inference"
configured_settings	{"allow_multiple_runs": true, "country_iso_code": "IND", "log_level": "error", "measure_power_secs": 1, "pue": 1.0, "save_to_api": false, "tracking_mode": "machine"}	{"allow_multiple_runs": true, "country_iso_code": "IND", "log_level": "error", "measure_power_secs": 1, "pue": 1.0, "save_to_api": false, "tracking_mode": "machine"}
hardware	{"cpu_model": "12th Gen Intel(R) Core(TM) i5-1235U", "gpu_count": 0, "gpu_model": "", "ram_total_size": 15.69796371459961}	{"cpu_model": "12th Gen Intel(R) Core(TM) i5-1235U", "gpu_count": 0, "gpu_model": "", "ram_total_size": 15.69796371459961}
backends	{"cpu": "unknown / not exposed by this version", "gpu": "unknown / not exposed by this version", "ram": "unknown / not exposed by this version"}	{"cpu": "unknown / not exposed by this version", "gpu": "unknown / not exposed by this version", "ram": "unknown / not exposed by this version"}
hardware_evidence	[{"class": "RAM", "mode": "unknown"}, {"class": "CPU", "mode": "windows_emi"}]	[{"class": "RAM", "mode": "unknown"}, {"class": "CPU", "mode": "windows_emi"}]

Question emissions change (%)

Relative to R1. Blue = lower, red = higher; grey = missing, failed, or zero reference. Q labels map to exact question text in the comparison table. Color saturates at +/-100%; cell numbers retain the actual change.

	R1	R2
Q1	0	-11.961
Q2	0	-8.0173

Largest changes: R2 versus R1

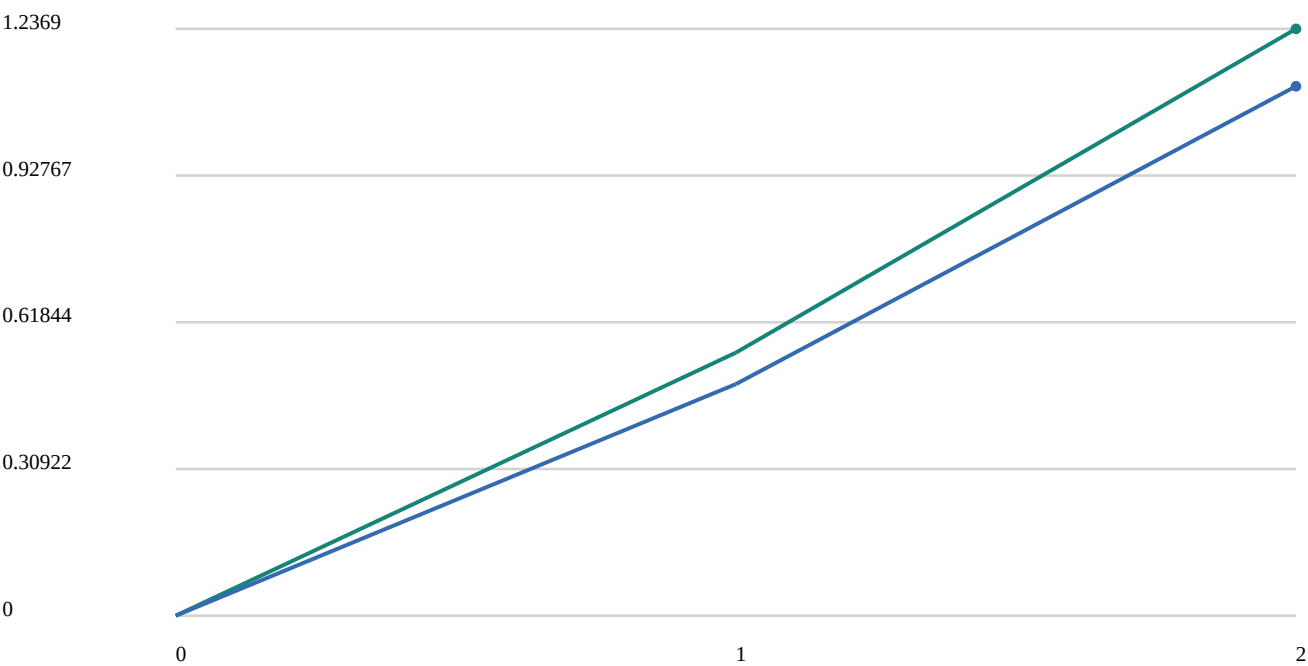
Absolute emissions changes for up to 12 matched successful questions. Left = decrease; right = increase. Labels show g CO2e.



Cumulative emissions through execution

X: number of questions executed. Y: cumulative g CO2e. Question identities at a given position can differ. Curves stop at the first missing measurement rather than implying zero cost. Failed-request measurements are included.

R1: 2/2 questions accumulated
R2: 2/2 questions accumulated



Aligned question comparison

Q identifiers follow the reference run. Text matching is exact. Mean emissions are calculated per question across all selected runs, using recorded values only; missing readings are excluded and the contributor count is shown. Recorded costs for failed attempts are included. Percentage changes are unavailable for a zero reference.

Q1: Repeated trials of the same operational task can cause agent pass rates to drop — propose a consistency-improvement approach appropriate for a local, resource-constrained LLM agent operating without external APIs.

Run	Position	Status	g CO2e	Mean g (n/runs)	Wh	Seconds	Delta g	Change %
R1	1	ok	0.55441	0.52125 (2/2)	0.77709	92.024	0	0
R2	1	ok	0.48809	0.52125 (2/2)	0.68414	82.275	-0.066311	-11.961

Q2: In a customer support setting, how should a business weigh the tradeoff between utilizing a heavy multi-agent LLM workflow that yields higher resolution accuracy, versus the reduced operational compute load of a smaller, less accurate standalone model?

Run	Position	Status	g CO2e	Mean g (n/runs)	Wh	Seconds	Delta g	Change %
R1	2	ok	0.68248	0.65512 (2/2)	0.95661	116.4	0	0
R2	2	ok	0.62777	0.65512 (2/2)	0.87991	109.22	-0.054716	-8.0173